Slicing Quantum Programs Where Developers Write Them: Static Slicing of Qiskit Source at Scale

Anonymous Author(s)

I. Introduction

[TODO: to be written]

II. Background and Related Work

[TODO: srcML and srcSlice, Behler et al. and QStatic, static analyzers for Qiskit, circuit tools; the light-cone rule stated and cited, not proved]

III. A Motivating Example

[TODO: a program of our own with computed qubit indices, loops and conditioned gates, e.g. `tool/tests/big/teleport_chain.py`]

IV. The Approach

[TODO: a semantic quantum layer over srcML (E1 to E5), over-approximation, circuit loops, one-pass slicing, OpenQASM front end]

V. Evaluation Design

[TODO: RQ1 faithfulness, RQ2 coverage, RQ3 scalability, RQ4 value for comprehension; all exploratory]

VI. Results

[TODO: to be written]

VII. Discussion and Threats to Validity

[TODO: to be written]

VIII. Conclusion

[TODO: to be written]

References